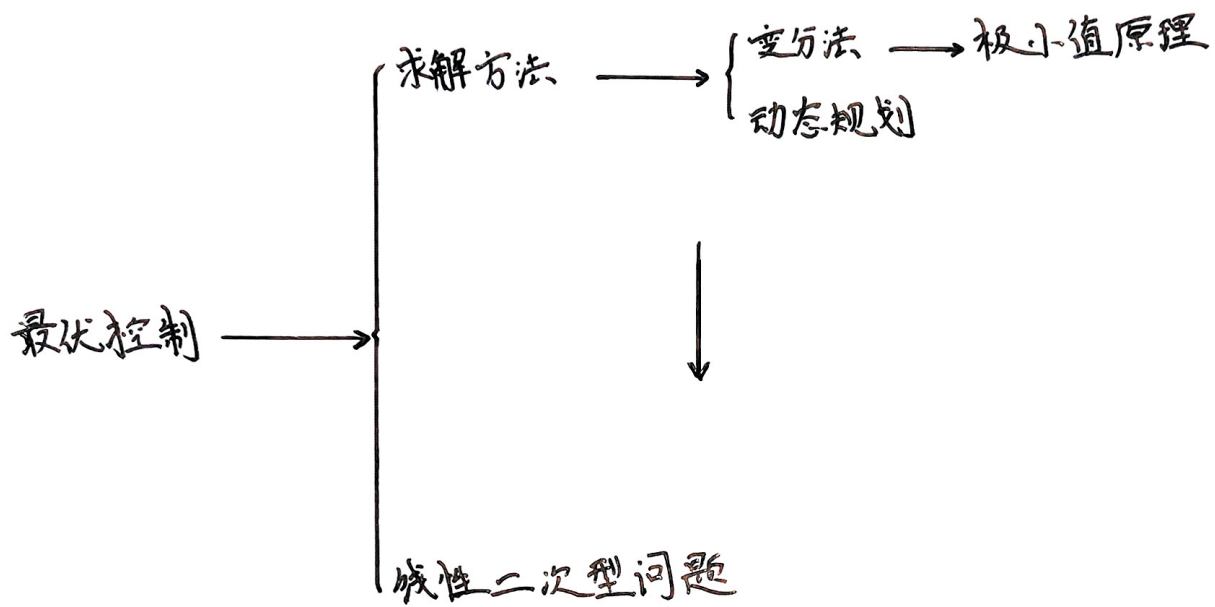
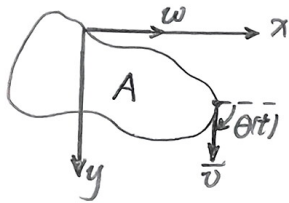




最优控制理论

第一章 最优控制概论

例1. 渔轮相对于海流的速度为 $\bar{v}$, 大小不变, 海流速度 $\bar{w}$ 方向大小一定, 求 $\theta(t)$ 的最优变化律, 使得在一定 t_f 内使得围得面积最大.



解 渔轮运动方程为

$$\begin{cases} \dot{x}(t) = v \cos \theta(t) + w \\ \dot{y}(t) = v \sin \theta(t) \end{cases}$$

初始条件与末端条件

$$\begin{cases} x(0) = x(t_f) = x_0 \\ y(0) = y(t_f) = y_0 \end{cases}$$

最优控制律, 性能指标选为面积:

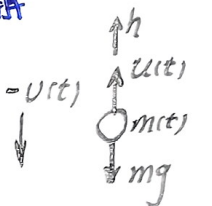
$$J = \oint y(t) dx(t) = \int_0^{t_f} y(t) \dot{x}(t) dt = \int_0^{t_f} y(t) (v \cos \theta(t) + w) dt$$

最大.

例2. 登月舱质量为 m , 高度为 h , 垂直速度为 v , 发动机推力为 u , 初始高度为 h_0 , 速度为 v_0 , 空载质量为 M , 燃料质量为 F .

求推力控制律使燃料消耗最小.

解



运动方程为

$$\begin{cases} \dot{h}(t) = v(t) \\ \dot{v}(t) = \frac{u(t)}{m(t)} - g \\ \dot{m}(t) = -k u(t) \end{cases}$$

初态与末态条件

$$\begin{cases} h(0) = h_0 \\ v(0) = v_0 \\ m(0) = M + F \end{cases} \quad \begin{cases} h(t_f) = 0 \\ v(t_f) = 0 \end{cases}$$

控制约束 $0 \leq u(t) \leq u_{\max}$

$J = m(t_f)$ 越大越好.

最优控制问题

最优控制问题的基本组成

① 系统数学模型 $\dot{x} = f(x(t), u(t), t), t \in [t_0, t_f]$

$x(t)$ 为状态向量, $u(t)$ 为控制向量.

② 边界条件与目标集

iii 初始时刻 t_0 及其初始状态 $x(t_0)$: $\rightarrow$ 已知.

iii 末端时刻 t_f 及其末端状态 $x(t_f)$: $\rightarrow$ 未知.

$\Downarrow$

$\varphi[x(t_f), t_f] = 0 \rightarrow$ $\begin{cases} x_f(t) \text{ 固定, } \varphi(\cdot) \text{ 仅有一点;} \\ x_f(t) \text{ 有约束, } \varphi(\cdot) \text{ 为 } r \text{ 维超曲面;} \\ x_f(t) \text{ 无约束, } \varphi(\cdot) \text{ 为 } r \text{ 维空间.} \end{cases}$

③ 容许控制 $\begin{cases} \text{范围受限的控制} \rightarrow \text{闭集} \\ \text{范围无限的控制} \rightarrow \text{开集} \end{cases} \Rightarrow \text{控制域} \Omega.$

④ 性能指标 — 代价函数 J .

· 最优控制的一般提法:

设系统状态方程及其初始条件为:

$$\begin{cases} \dot{x}(t) = f(x(t), u(t), t) \\ x(t_0) = x_0 \end{cases}$$

要求确定最优控制 $u^*(t)$, 使系统从已知初态 $x(t_0)$ 转移到末态 $x(t_f)$.

使性能指标:

$$J = \varphi(x(t_f), t_f) + \int_{t_0}^{t_f} L(x(t), u(t), t) dt$$

有极值, 同时满足:

ii) 控制约束: $g(x(t), u(t), t) \geq 0$;

iii) 目标约束: $\varphi(x(t_f), t_f) = 0$.

性能指标类型

Ⅰ 积分型指标 $J = \int_{t_0}^{t_f} L(x(t), u(t), t) dt$

→ 拉格朗日问题.

ii) 最短时间控制 $J = \int_{t_0}^{t_f} dt = t_f - t_0$ ($L = 1$)

iii) 最小能耗控制 $J = \int_{t_0}^{t_f} \sum |u_j(t)| dt$ ($L = \sum |u_j(t)|$)

iiii) 最小能量控制 $J = \int_{t_0}^{t_f} u^T(t) u(t) dt$ ($L = u^T(t) u(t)$)

Ⅱ 末值型指标 $J = \varphi(x(t_f), t_f)$

→ 迈耶问题.

Ⅲ 复合型指标

→ 波尔扎问题.

ii) 有限时间线性状态调节器:

iii) 有限时间线性跟踪系统.

第二章 变分法与极小值原理

1. 变分法基本原理

注记 对于某些函数中的所有函数 $x(t)$, J 有唯一确定值与之对应, 则 J 为依赖于 $x(t)$ 的泛函, 记为 $J = J[x(t)]$

例如, 积分是一种泛函: 令 $J = \int_0^1 x(t) dt$

$$x(t) = t \text{ 时, } J(x) = \frac{1}{2}$$

$$x(t) = \sin t \text{ 时, } J(x) = \sin 1$$

• R^n 为 n 维线性赋范空间, R 为实数集. 若有

$$y = J(x), \forall x \in R^n, y \in R$$

称 $J(x)$ 为 R^n 到 R 的泛函算子.

• 若 $J(x_1 + x_2) = J(x_1) + J(x_2)$, $J(\alpha x_1) = \alpha J(x_1)$, 则 J 为线性泛函算子 (如积分)

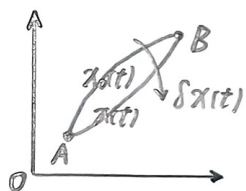
• **泛函的连续性** J 为 R^n 子集 $D(J)$ 到 R 中的泛函算子:

若对每一个收敛到 $x_0 \in D(J)$ 的序列 $x_n \in D(J)$, 有 $\lim_{n \rightarrow \infty} J(x_n) = J(x_0)$, 则 $J(x)$ 在 x_0 处连续.

• 线性泛函必然连续.

现在, 泛函的映射关系是由函数映射到一个值, 因此类似于求函数的极小值, 接下来讨论如何求泛函的极小值.

• 函数的变分 (自变量增量): $x(t)$ 的变分为: $\delta x = x - x_0$



δx 可引起 $J(x)$ 变化: $J(x + \varepsilon \delta x) \begin{cases} \varepsilon = 0 \text{ 为原泛函 } J(x) \\ \varepsilon = 1 \text{ 为增加后泛函 } J(x + \delta x) \end{cases}$

• 泛函的变分 (泛函增量): $J(x)$ 为 R^n 上的线性连续泛函, 若

$$\Delta J(x) = J(x + \delta x) - J(x) = \underbrace{L(x, \delta x)}_{\delta x \text{ 的线性连续泛函}} + \underbrace{r(x, \delta x)}_{\text{高阶无穷小}}$$

则 δJ 为 $J(x)$ 的变分.

连续泛函的变分定理 $J(x)$ 为 R^n 上的连续泛函, 若 $x = x_0$ 处 $J(x)$ 可微, 则 $J(x)$ 在 x_0 处的变分为:

$$\delta J(x_0, \delta x) = \left. \frac{\partial}{\partial \varepsilon} J(x_0, \varepsilon \delta x) \right|_{\varepsilon=0}$$

• 若 $J = \int_{t_0}^{t_f} L(x, \dot{x}, t) dt$, 则 $\delta J = \int_{t_0}^{t_f} \left(\frac{\partial L}{\partial x} \delta x + \frac{\partial L}{\partial \dot{x}} \delta \dot{x} \right) dt$

泛函的极值 $J(x)$ 为 R^n 某子集 D 上的线性连续泛函, $x_0 \in D$, 若存在 $\delta > 0$, 使

$$U(x_0, \delta) = \{x \mid \|x - x_0\| < \delta, x \in R^n\}$$

在 $x \in U(x_0, \delta)$ 时, 有

$$\begin{cases} \Delta J(x) = J(x) - J(x_0) \geq 0 \rightarrow J \text{ 在 } x_0 \text{ 处有极小值.} \\ \Delta J(x) = J(x) - J(x_0) \leq 0 \rightarrow J \text{ 在 } x_0 \text{ 处有极大值.} \end{cases}$$

泛函极值必要条件: $J(x)$ 在 x_0 处有极值 $\Rightarrow \delta J(x_0, \delta x) = 0$

泛函极值充要条件: $\begin{cases} \delta J(x_0, \delta x) = 0, \delta^2 J(x_0, \delta x) > 0 \Leftrightarrow J(x) \text{ 在 } x_0 \text{ 有极小值.} \\ \delta J(x_0, \delta x) = 0, \delta^2 J(x_0, \delta x) < 0 \Leftrightarrow J(x) \text{ 在 } x_0 \text{ 有极大值.} \end{cases}$

变分法求泛函极值 讨论泛函 $J(x) = \int_{t_0}^{t_f} L(x, \dot{x}, t) dt$.

两端固定问题 $x(t_0) = x_0, x(t_f) = x_f$

无约束极值 $\min_x J(x) = \int_{t_0}^{t_f} L(x, \dot{x}, t) dt$

$\Downarrow$

极值必要条件 $\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = 0$ (Euler 方程)

$$L(x^* + \delta x, \dot{x}^* + \delta \dot{x}, t) = L(x^*, \dot{x}^*, t) + \left. \frac{\partial L}{\partial x} \right|_{x^*} \delta x + \left. \frac{\partial L}{\partial \dot{x}} \right|_{\dot{x}^*} \delta \dot{x} + \dots \quad (\text{Taylor 展开})$$

$$\Rightarrow \delta J = \int_{t_0}^{t_f} \left(\frac{\partial L}{\partial x} \delta x + \frac{\partial L}{\partial \dot{x}} \delta \dot{x} \right) dt$$

对第二项用分部积分: $\delta \dot{x} = \dot{x}(t) - \dot{x}_0(t), \frac{d}{dt} \delta x = \frac{d}{dt} (x(t) - x_0(t)) = \delta \dot{x}$

$$\int_{t_0}^{t_f} \frac{\partial L}{\partial \dot{x}} \delta \dot{x} dt = \left. \frac{\partial L}{\partial \dot{x}} \delta x \right|_{t_0}^{t_f} - \int_{t_0}^{t_f} \left(\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right) \delta x dt$$

$$\Rightarrow \delta J = \int_{t_0}^{t_f} \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right) \delta x dt + \left. \frac{\partial L}{\partial \dot{x}} \delta x \right|_{t_0}^{t_f} = 0$$

由变分定理: 由于两端固定, $\delta x(t_0) = \delta x(t_f) = 0$ 则必有

$$\begin{aligned} &\Downarrow \\ f(x) \text{ 和任意 } \eta(x) \text{ 正交} &\left\{ \begin{aligned} \frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} &= 0 \quad (\delta x \text{ 任意}) \\ \frac{\partial L}{\partial \dot{x}} \Big|_{t_f} \delta x(t_f) &= \frac{\partial L}{\partial \dot{x}} \Big|_{t_0} \delta x(t_0) \quad (\text{两端固定必然}) \end{aligned} \right. \\ &\Downarrow \\ f(x) &= 0 \end{aligned}$$

例1. $J(x) = \int_0^{\frac{\pi}{2}} (\dot{x}(t) - x(t)) dt, x(0) = 0, x(\frac{\pi}{2}) = 2$, 求 $\min_x J$ 时的 x .

解 $L(\dot{x}, x, t) = \dot{x}^2 - x^2 \quad \frac{\partial L}{\partial x} = -2x, \frac{\partial L}{\partial \dot{x}} = 2\dot{x}$

由 Euler 方程 $\ddot{x} + x = 0$, 则 $x = 2 \sin t$.

有约束极值 $\begin{cases} \min_x J(x) = \int_{t_0}^{t_f} g(x, \dot{x}, t) dt \\ f(x, \dot{x}, t) = 0 \end{cases}$

$\Downarrow$

极值必要条件: $\begin{cases} \frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = 0 \\ L(x, \dot{x}, \lambda, t) = g(x, \dot{x}, t) + \lambda(t) f(x, \dot{x}, t) \end{cases}$

例2. 对 $\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u \rightarrow$ 系统方程为约束.

取 $J = \frac{1}{2} \int_0^2 u^2(t) dt$, $x(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $x(2) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, 求 J 取极值的 x .

解 $g = \frac{1}{2} u^2$, $f = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u - \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix}$

$$L(x, u, \lambda, t) = \frac{1}{2} u^2 + \lambda_1 (x_2 - \dot{x}_1) + \lambda_2 (u - \dot{x}_2)$$

$$\Rightarrow \begin{cases} \frac{\partial L}{\partial x_1} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_1} = \lambda_1 = 0 \\ \frac{\partial L}{\partial x_2} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_2} = \lambda_1 + \lambda_2 = 0 \\ \frac{\partial L}{\partial u} - \frac{d}{dt} \frac{\partial L}{\partial \dot{u}} = u + \lambda_2 = 0 \end{cases} \Rightarrow \begin{cases} \lambda_1 = a \\ \lambda_2 = -at + b \\ u = at - b \end{cases}$$

$$\text{由状态方程} \Rightarrow \begin{cases} \dot{x}_1 = \frac{1}{6} at^3 - \frac{1}{2} bt^2 + ct + d \\ \dot{x}_2 = \frac{1}{2} at^2 - bt + c \end{cases}$$

$\downarrow$

$$\begin{cases} x_1 = \frac{1}{24} at^4 - \frac{7}{24} bt^3 + ct^2 + dt \\ x_2 = \frac{1}{6} at^3 - \frac{7}{24} bt^2 + ct + d \end{cases}$$

注: L 中的自变量包含状态 x 和输入 u .

J 取极小值的充分条件 (Legendre 条件)

$$\textcircled{1} \begin{pmatrix} \frac{\partial^2 L}{\partial x^2} & \frac{\partial^2 L}{\partial x \partial \dot{x}} \\ \frac{\partial^2 L}{\partial x \partial \dot{x}} & \frac{\partial^2 L}{\partial \dot{x}^2} \end{pmatrix} \text{正定} \quad \textcircled{2} \frac{\partial^2 L}{\partial \dot{x}^2} - \frac{d}{dt} \frac{\partial^2 L}{\partial x \partial \dot{x}} \geq 0 \text{ 且 } \frac{\partial^2 L}{\partial \dot{x}^2} > 0$$

$$\textcircled{3} \frac{\partial^2 L}{\partial x^2} - \frac{d}{dt} \frac{\partial^2 L}{\partial x \partial \dot{x}} > 0 \text{ 且 } \frac{\partial^2 L}{\partial \dot{x}^2} \geq 0$$

横截条件 [J 极值: Euler 方程 + 横截条件]

$$\text{① 在固定横截条件} \quad \frac{\partial L}{\partial \dot{x}} \Big|_{t_f} \delta x(t_f) = \frac{\partial L}{\partial \dot{x}} \Big|_{t_0} \delta x(t_0)$$

固定起点和终点

$$x(t_0) = x_0, x(t_f) = x_f$$

固定起点和自由终点

$$x(t_0) = x_0, \frac{\partial L}{\partial \dot{x}} \Big|_{t_f} = 0$$

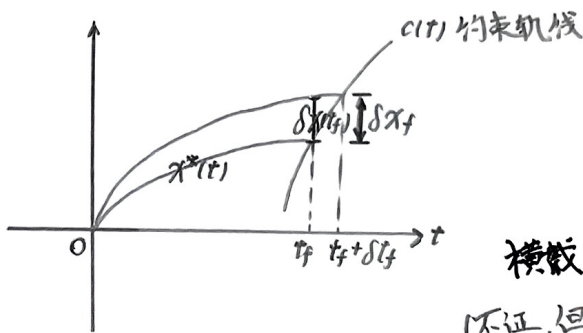
自由起点和固定终点

$$\frac{\partial L}{\partial \dot{x}} \Big|_{t_0} = 0, x(t_f) = x_f$$

自由起点和自由终点

$$\frac{\partial L}{\partial \dot{x}} \Big|_{t_0} = 0, \frac{\partial L}{\partial \dot{x}} \Big|_{t_f} = 0$$

② t_f 自由横截条件



$$\begin{cases} \text{末端有约束: } \begin{cases} \delta x(t_f) = \delta x_f - \dot{x}(t_f) \delta t_f \\ \delta x_f = \dot{c}(t_f) \delta t_f \end{cases} \quad (\text{近似}) \\ \text{末端无约束: } \delta x(t_f) = \delta x_f - \dot{x}(t_f) \delta t_f \end{cases}$$

$$\text{横截条件: } \frac{\partial L}{\partial \dot{x}} \Big|_{t_f} \delta x(t_f) - \frac{\partial L}{\partial \dot{x}} \Big|_{t_0} \delta x(t_0) + L(x^*, \dot{x}^*, t) \Big|_{t_f} \delta t_f = 0$$

(不证, 但方法一致) ↓
横截条件

$$\text{末端自由} \begin{cases} \frac{\partial L}{\partial \dot{x}} \Big|_{t_f} = 0 \\ x(t_0) = x_0 \\ (L - \dot{x} \frac{\partial L}{\partial \dot{x}}) \Big|_{t_f} = 0 \end{cases}$$

$$\text{末端约束 } c(t) \begin{cases} x(t_f) = c(t_f) \\ x(t_0) = x_0 \\ (L + (\dot{c} - \dot{x}) \frac{\partial L}{\partial \dot{x}}) \Big|_{t_f} = 0 \end{cases}$$

例3. $t_0 = 0$, $x(0) = 1$, $x(t_f) = 2 - t_f$, $J = \int_0^{t_f} (1 + \dot{x}^2)^{\frac{1}{2}} dt$, 求 $x^*(t)$ 及 t_f .

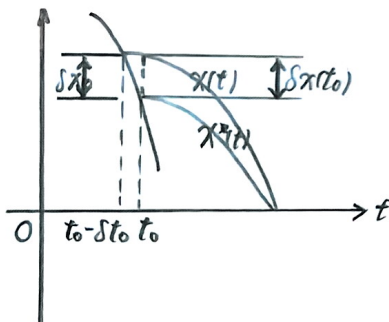
解 $L(x, \dot{x}, t) = (1 + \dot{x}^2)^{\frac{1}{2}}$, $c(t) = 2 - t$

$$\Rightarrow \text{Euler 方程: } \frac{d}{dt} \left(\frac{\dot{x}}{\sqrt{1 + \dot{x}^2}} \right) = 0 \Rightarrow x(t) = at + b$$

$$\text{由横截条件, 可得 } a = 1, b = 1. \Rightarrow x^*(t) = t + 1$$

$$x(t_f) = c(t_f) \Rightarrow t_f^* = \frac{1}{2}$$

③ t_0 自由横截条件



$$\begin{cases} \text{首端有约束: } \begin{cases} \delta x(t_0) = \delta x_0 - \dot{x}(t_0) \delta t_0 \\ \delta x_0 = -\dot{\varphi}(t_0) \delta t_0 \end{cases} \\ \text{首端无约束: } \delta x(t_0) = \delta x_0 + \dot{x}(t_0) \delta t_0 \end{cases}$$

$$\text{横截条件: } \frac{\partial L}{\partial \dot{x}} \Big|_{t_f} \delta x(t_f) - \frac{\partial L}{\partial \dot{x}} \Big|_{t_0} \delta x(t_0) + L(x^* + \delta x, \dot{x}^* + \delta \dot{x}, t) \Big|_{t_0} \delta t_0 = 0$$

$$\text{首端自由} \begin{cases} \frac{\partial L}{\partial \dot{x}} \Big|_{t_0} = 0 \\ x(t_f) = x_f \\ (L - \dot{x} \frac{\partial L}{\partial \dot{x}}) \Big|_{t_0} = 0 \end{cases}$$

$$\text{首端约束 } \varphi(t) \begin{cases} x(t_0) = \varphi(t_0) \\ x(t_f) = x_f \\ (L - (\dot{\varphi} - \dot{x}) \frac{\partial L}{\partial \dot{x}}) \Big|_{t_0} = 0 \end{cases}$$

2. 变方法与最优控制 极小值原理

连续系统最优控制问题 系统状态 $\dot{x}(t) = f(x(t), u(t), t)$

$$x(t_0) = x_0$$

$$\text{性能泛函 } J(x) = \underbrace{\varphi(x(t_f), t_f)}_{\text{末态约束}} + \underbrace{\int_{t_0}^{t_f} L(x, u, t) dt}_{\text{过程约束}}$$

$$\text{目标集 } \varphi(x(t_f), t_f) = 0$$

简单的方法是从系统方程中解出 u , 代入 L 后用 Euler 方程和横截条件解 x^* 和 u^*
但有时 u 难以求出, 必须寻找其他方法.

II 末端时刻固定问题

$$\begin{cases} \dot{x}(t) = f(x(t), u(t), t) \\ x(t_0) = x_0 \\ J(x) = \varphi(x(t_f)) + \int_{t_0}^{t_f} L(x, u, t) dt \\ \varphi(x(t_f)) = 0 \end{cases}$$

• 最优解必要条件 (不证)

① 末端状态是约束 $\varphi(x(t_f)) = 0$

i) 哈密顿函数 $H(x, u, \lambda, t) = L(x, u, t) + \lambda^T(t) f(x, u, t)$

满足正则方程
$$\begin{cases} \dot{x}(t) = \frac{\partial H}{\partial \lambda} \\ \dot{\lambda}(t) = -\frac{\partial H}{\partial x} \quad (\text{协态方程}) \end{cases}$$

ii) 边界条件
$$\begin{cases} x(t_0) = x_0 \\ \varphi(x(t_f)) = 0 \\ \lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)} + \frac{\partial \varphi}{\partial \lambda(t_f)} \gamma(t_f) \end{cases}$$

iii) 极值条件: $\frac{\partial H}{\partial u} = 0$

② 末端自由

i) 哈密顿函数满足正则方程.

ii) 边界条件
$$\begin{cases} x(t_0) = x_0 \\ \lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)} \end{cases}$$

iii) 极值条件

③ 末端状态固定 $x(t_f) = x_f$

i) 系统完全可控;

ii) 哈密顿函数满足正则方程;

iii) 边界条件
$$\begin{cases} x(t_0) = x_0 \\ x(t_f) = x_f \end{cases}$$

iv) 极值条件.

例1.
$$\begin{cases} \dot{x}(t) = u(t) \\ x(t_0) = x_0 \\ J = \frac{1}{2}c x^2(t_f) + \frac{1}{2} \int_{t_0}^{t_f} u^2(t) dt \end{cases}$$

t_0, t_f 给定, $c > 0$, $x(t_f)$ 自由, 求 J 最小时的 u .

解 $f = u$, $L = \frac{1}{2}u^2$, $\varphi = \frac{1}{2}c x^2(t_f)$

$H(x, u, \lambda, t) = \frac{1}{2}u^2 + \lambda u$

$$\begin{cases} u + \lambda = 0 & \Rightarrow u = -\lambda = -c x(t_f) \\ u = \dot{x} \\ \dot{\lambda} = 0 \end{cases} \Rightarrow \dot{x} = -c x(t_f)$$

$$\lambda(t_f) - \frac{\partial \varphi}{\partial x(t_f)} = c x(t_f)$$

$\Rightarrow x = x_0 - c x(t_f)(t - t_0) \Rightarrow u = -\frac{c x_0}{1 + c(t_f - t_0)}$

例2.
$$\begin{cases} \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u(t) \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}(0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ J = \frac{1}{2} \int_0^1 u^2 dt \end{cases}$$

$t_f = 1$, $x_1(t_f) + x_2(t_f) = 1$, 求 J 最小时的 u .

解 $\varphi(x(t_f)) = 0$, $L = \frac{1}{2}u^2$, $f_1 = x_2(t)$, $f_2 = u(t)$, $\varphi = x_1(1) + x_2(1) - 1$

$H = \frac{1}{2}u^2 + \lambda_1 x_2 + \lambda_2 u$

$$\begin{cases} \dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = 0 \\ \dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = -\lambda_1 \end{cases} \Rightarrow \begin{cases} \lambda_1 = c_1 \\ \lambda_2 = -c_1 t + c_2 \\ u = c_1 t - c_2 \end{cases} \Rightarrow \begin{cases} x_1 = \frac{1}{6}c_1 t^3 - \frac{1}{2}c_2 t^2 + c_3 t + c_4 \\ x_2 = \frac{1}{2}c_1 t^2 - c_2 t + c_3 \end{cases}$$

$$\begin{cases} x_1(0) = x_2(0) = 0 \\ \lambda_1(1) = \frac{\partial \varphi}{\partial x_1(1)} = 1 \\ \lambda_2(1) = \frac{\partial \varphi}{\partial x_2(1)} = 1 \end{cases}$$

$\Downarrow$
 $c_3 = c_4 = 0, c_1 = -\frac{3}{7}, c_2 = -\frac{6}{7}$
 $u = -\frac{3}{7}(t-2)$

$\frac{\partial H}{\partial u} = u + \lambda_2 = 0$

• 最优解充要条件:
$$\begin{cases} \theta(x(t_f), x) = \varphi(x(t_f)) + \gamma(t) \varphi(x(t_f)) \\ H(x, u, \lambda, t) = L(x, u, t) + \lambda f(x, u, t) \end{cases}$$

① $\begin{pmatrix} \frac{\partial^2 H}{\partial x^2} & \frac{\partial^2 H}{\partial x \partial u} \\ \frac{\partial^2 H}{\partial u \partial x} & \frac{\partial^2 H}{\partial u^2} \end{pmatrix}$ 正定 且 $\frac{\partial^2 \theta}{\partial x(t_f)^2} \geq 0$ 或 ② $\begin{pmatrix} \frac{\partial^2 H}{\partial x^2} & \frac{\partial^2 H}{\partial x \partial u} \\ \frac{\partial^2 H}{\partial u \partial x} & \frac{\partial^2 H}{\partial u^2} \end{pmatrix}$ 半正定 且 $\frac{\partial^2 \theta}{\partial x(t_f)^2} > 0$.

② 末端时刻自由的解

$$\begin{cases} \dot{x} = f(x, u, t) \\ x(t_0) = x_0 \\ J = \varphi(x(t_f), t_f) + \int_{t_0}^{t_f} L(x, u, t) dt \\ \psi(x(t_f), t_f) = 0 \end{cases}$$

① 末端状态受约束

(i) 哈密顿函数满足正则方程.

$$\text{iii) 边界条件} \begin{cases} x(t_0) = x_0 \\ \psi(x(t_f), t_f) = 0 \\ \lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)} + \frac{\partial \psi}{\partial x(t_f)} \gamma(t_f) \end{cases}$$

iii) 极值条件

$$\text{iv) 末端条件: } H(t_f) = -\frac{\partial \varphi}{\partial t_f} - \gamma(t_f) \frac{\partial \psi}{\partial t_f}$$

② 末端自由

(i) 哈密顿函数满足正则方程.

$$\text{iii) 边界条件} \begin{cases} x(t_0) = x_0 \\ \lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)} \end{cases}$$

iii) 极值条件

$$\text{iv) 末端条件: } H(t_f) = -\frac{\partial \varphi}{\partial t_f}$$

③ 末端固定

(i) 哈密顿函数满足正则方程.

$$\text{iii) 边界条件} \begin{cases} x(t_0) = x_0 \\ x(t_f) = x_f \end{cases}$$

iii) 极值条件

$$\text{iv) 末端条件: } H(t_f) = -\frac{\partial \varphi}{\partial t_f}$$

④ 角点条件与内点约束: $x(t)$ 连续但不可微

• 角点处函数极值: 对分段连续函数 $J(x)$

(i) 满足 Euler 方程;

$$\text{iii) } \begin{cases} \frac{\partial L}{\partial \dot{x}}|_{t_i^-} = \frac{\partial L}{\partial \dot{x}}|_{t_i^+} \\ \left(L - \dot{x} \frac{\partial L}{\partial \dot{x}} \right)_{t_i^-} = - \left(L - \dot{x} \frac{\partial L}{\partial \dot{x}} \right)_{t_i^+} \end{cases} \Rightarrow \text{有极值.}$$

• 内点约束的最优解

$$\begin{cases} \dot{x} = f(x, u, t) \\ x(0) = x_0 \\ J = \varphi(x(t_f)) + \int_{t_0}^{t_f} L(x, u, t) dt \\ \eta(x(t_1), t_1) = 0 \quad \text{—— 内点约束} \end{cases}$$

(i) 哈密顿函数满足正则方程,

$$\text{iii) 边界条件} \begin{cases} x(t_0) = x_0 \\ \lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)} \end{cases}$$

iii) 极值条件

iv) 内点约束

$$\begin{cases} n(x(t_i), t_i) = 0 \\ \lambda(t_i^-) = \lambda(t_i^+) + \frac{\partial n}{\partial x(t_i)} \pi \\ H(t_i^-) = H(t_i^+) - \pi \frac{\partial n}{\partial t_i} \end{cases}$$

连续系统的极小值原理 (不证)

古典变分法: u 不受约束
极小值原理: u 受约束

表 3-1 定常系统极小值原理

末端时刻	性能指标	末端状态	正则方程	极小值条件	边界条件	H 变化律		
t_f 固定	$J = \varphi[x(t_f)]$	末端自由	$\dot{x} = \frac{\partial H}{\partial \lambda}$ $\dot{\lambda} = -\frac{\partial H}{\partial x}$	$H^* = H(x^*, \lambda, u^*) = \min_{u \in U} H(x^*, \lambda, u)$	$x(t_0) = x_0$ $\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$	$H^*(t) = H^*(t_f)$ $= \text{const}$		
		末端约束	式中 $H = \lambda^T(t)f(x, u)$		$x(t_0) = x_0, \psi[x(t_f)] = 0$ $\lambda(t_f) = \left[\frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} \right]_{t_f}$	$H^*(t) = H^*(t_f)$ $= \text{const}$		
	$J = \int_{t_0}^{t_f} L(x, u) dt$	末端固定	$\dot{x} = \frac{\partial H}{\partial \lambda}$ $\dot{\lambda} = -\frac{\partial H}{\partial x}$ 式中 $H = L(x, u) + \lambda^T(t)f(x, u)$		$x(t_0) = x_0, x(t_f) = x_f$ ($\lambda(t_f)$ 未知)	$H^*(t) = H^*(t_f)$ $= \text{const}$		
		末端自由			$x(t_0) = x_0$ $\lambda(t_f) = 0$			
		末端约束			$x(t_0) = x_0, \psi[x(t_f)] = 0$ $\lambda(t_f) = \left[\frac{\partial \psi^T}{\partial x} \right]_{t_f}$			
	$J = \varphi[x(t_f)] + \int_{t_0}^{t_f} L(x, u) dt$	末端自由	$H = L(x, u) + \lambda^T(t)f(x, u)$		$x(t_0) = x_0$ $\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$	$H^*(t) = H^*(t_f)$ $= \text{const}$		
		末端约束			$x(t_0) = x_0, \psi[x(t_f)] = 0$ $\lambda(t_f) = \left[\frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} \right]_{t_f}$			
	t_f 自由	$J = \varphi[x(t_f)]$	末端自由		$\dot{x} = \frac{\partial H}{\partial \lambda}$ $\dot{\lambda} = -\frac{\partial H}{\partial x}$	$H^* = H(x^*, \lambda, u^*) = \min_{u \in U} H(x^*, \lambda, u)$	$x(t_0) = x_0$ $\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$	$H^*(t_f) = 0$
			末端约束		式中 $H = \lambda^T(t)f(x, u)$		$x(t_0) = x_0, \psi[x(t_f)] = 0$ $\lambda(t_f) = \left[\frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} \right]_{t_f}$	$H^*(t_f) = 0$
		$J = \int_{t_0}^{t_f} L(x, u) dt$	末端固定		$\dot{x} = \frac{\partial H}{\partial \lambda}$ $\dot{\lambda} = -\frac{\partial H}{\partial x}$ 式中 $H = L(x, u) + \lambda^T(t)f(x, u)$		$x(t_0) = x_0, x(t_f) = x_f$ ($\lambda(t_f)$ 未知)	$H^*(t_f) = 0$
末端自由			$x(t_0) = x_0$ $\lambda(t_f) = 0$					
末端约束			$x(t_0) = x_0, \psi[x(t_f)] = 0$ $\lambda(t_f) = \left[\frac{\partial \psi^T}{\partial x} \right]_{t_f}$					
$J = \varphi[x(t_f)] + \int_{t_0}^{t_f} L(x, u) dt$		末端自由	$H = L(x, u) + \lambda^T(t)f(x, u)$	$x(t_0) = x_0$ $\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$	$H^*(t_f) = 0$			
		末端约束		$x(t_0) = x_0, \psi[x(t_f)] = 0$ $\lambda(t_f) = \left[\frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} \right]_{t_f}$				

表 3-2 时变系统极小值原理

末端时刻	性能指标	末端状态	正则方程	极小值条件	边界条件	H 变化律
t_f 固定	$J = \varphi[x(t_f), t_f]$	末端自由	$\dot{x} = \frac{\partial H}{\partial x}$	$H^* = H(x^*, \lambda^*, t) = \min_{u \in U} H(x, u, t)$	$x(t_0) = x_0$	$H^*(t)$
		末端约束	$\dot{\lambda} = -\frac{\partial H}{\partial x}$ 式中 $H = \lambda^T(t)f(x, u, t)$		$\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$ $x(t_0) = x_0, \varphi[x(t_f), t_f] = 0$ $\lambda(t_f) = \left[\frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$	$= H^*(t_f)$ $-\int_{t_0}^{t_f} \frac{\partial H}{\partial t} dt$
	$J = \int_{t_0}^{t_f} L(x, u, t) dt$	末端固定			$x(t_0) = x_0, x(t_f) = x_f$ ($\lambda(t_f)$ 未知)	$H^*(t)$
		末端自由	$\dot{x} = \frac{\partial H}{\partial x}$		$x(t_0) = x_0$ $\lambda(t_f) = 0$	$= H^*(t_f)$
		末端约束	$\dot{\lambda} = -\frac{\partial H}{\partial x}$ 式中 $H = L(x, u, t)$		$x(t_0) = x_0, \varphi[x(t_f), t_f] = 0$ $\lambda(t_f) = \frac{\partial \psi^T}{\partial x(t_f)} v$	$-\int_{t_0}^{t_f} \frac{\partial H}{\partial t} dt$
	$J = \varphi[x(t_f), t_f] + \int_{t_0}^{t_f} L(x, u, t) dt$	末端自由	$H = L(x, u, t) + \lambda^T(t)f(x, u, t)$		$x(t_0) = x_0$ $\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$	$H^*(t)$
t_f 自由	$J = \varphi[x(t_f), t_f]$	末端自由	$\dot{x} = \frac{\partial H}{\partial x}$	$H^* = H(x^*, \lambda^*, t) = \min_{u \in U} H(x, u, t)$	$x(t_0) = x_0$ $\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$	$H^*(t_f) = -\frac{\partial \varphi}{\partial t_f}$
		末端约束	$\dot{\lambda} = -\frac{\partial H}{\partial x}$ 式中 $H = \lambda^T(t)f(x, u, t)$		$x(t_0) = x_0, \varphi[x(t_f), t_f] = 0$ $\lambda(t_f) = \left[\frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$	$H^*(t_f) = -\left[\frac{\partial \varphi}{\partial t} + v^T \frac{\partial \psi}{\partial t} \right]_{t_f}$
	$J = \int_{t_0}^{t_f} L(x, u, t) dt$	末端固定			$x(t_0) = x_0, x(t_f) = x_f$ ($\lambda(t_f)$ 未知)	$H^*(t_f) = 0$
		末端自由	$\dot{x} = \frac{\partial H}{\partial x}$		$x(t_0) = x_0$ $\lambda(t_f) = 0$	$H^*(t_f) = 0$
		末端约束	$\dot{\lambda} = -\frac{\partial H}{\partial x}$ 式中 $H = L(x, u, t)$		$x(t_0) = x_0, \varphi[x(t_f), t_f] = 0$ $\lambda(t_f) = \frac{\partial \psi^T}{\partial x(t_f)} v$	$H^*(t_f) = -v^T \frac{\partial \psi}{\partial t_f}$
	$J = \varphi[x(t_f), t_f] + \int_{t_0}^{t_f} L(x, u, t) dt$	末端自由	$H = L(x, u, t) + \lambda^T(t)f(x, u, t)$		$x(t_0) = x_0$ $\lambda(t_f) = \frac{\partial \varphi}{\partial x(t_f)}$	$H^*(t_f) = -\frac{\partial \varphi}{\partial t_f}$
		末端约束			$x(t_0) = x_0, \varphi[x(t_f), t_f] = 0$ $\lambda(t_f) = \left[\frac{\partial \varphi}{\partial x} + \frac{\partial \psi^T}{\partial x} v \right]_{t_f}$	$H^*(t_f) = -\left[\frac{\partial \varphi}{\partial t} + v^T \frac{\partial \psi}{\partial t} \right]_{t_f}$

• 极小值原理的特点：
 (i) 容许控制条件放宽；
 (ii) 使 H 全局最小；
 (iii) 不要求 H 对控制的可微性（极值条件被修改）

例 3.
$$\begin{cases} \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u(t) \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{cases}$$

$|u(t)| \leq 1$, 末端状态自由且 $J = x_2(1)$. 求 u .

解 $\varphi(x(t_f)) = x_2(1), t_f = 1$

$$H = \lambda_1(-x_1 + u) + \lambda_2 x_1$$

$$\begin{cases} \dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = \lambda_1 - \lambda_2 \\ \dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = 0 \end{cases} \Rightarrow \begin{cases} \lambda_1 = c_1 e^t + c_2 \\ \lambda_2 = c_2 \end{cases}$$

$$\begin{cases} \lambda_1(1) = \frac{\partial \varphi}{\partial x_1(1)} = 0 \\ \lambda_2(1) = \frac{\partial \varphi}{\partial x_2(1)} = 1 \end{cases} \Rightarrow \begin{cases} \lambda_1 = 1 - e^{t-1} \\ \lambda_2 = 1 \end{cases}$$

取 u 使 H 最小, 关注 λ_1, u 项, 由 $\lambda_1(t) > 0$, 则 $u = -1$ 即可.

离散系统最优控制问题 $\begin{cases} x(n+1) = f(x(n), u(n), n) \\ J = \sum_{n=0}^{N-1} L(x(n), u(n), n) \end{cases}$

• J 取最值的必要条件: 令 $L_n = L + \lambda(n+1)(f - x(n+1))$

i) 横截条件: $\frac{\partial L_n}{\partial x(n)} \Big|_N \delta x(N) = \frac{\partial L_{n-1}}{\partial x(n)} \Big|_0 \delta x(0)$

ii) 离散 Euler 方程 $\begin{cases} \frac{\partial L_n}{\partial x(n)} + \frac{\partial L_{n-1}}{\partial x(n)} = 0 \\ \frac{\partial L_n}{\partial u(n)} = 0 \end{cases}$

例4. $\begin{cases} x(n+1) = x(n) + u(n) \\ x(0) = 1, x(5) = 0 \\ J = \frac{1}{2} \sum_{n=0}^4 u^2(n) \end{cases}$

求 $u(n)$.

解 $\begin{cases} L_n = \frac{1}{2} u^2(n) + \lambda(n+1)(-x(n+1) + x(n) + u(n)) \\ L_{n-1} = \frac{1}{2} u^2(n-1) + \lambda(n)(-x(n) + x(n-1) + u(n-1)) \end{cases}$

$$\frac{\partial L_n}{\partial x(n)} = \lambda(n+1), \frac{\partial L_{n-1}}{\partial x(n)} = -\lambda(n), \frac{\partial L_n}{\partial u(n)} = u(n) + \lambda(n+1).$$

由 Euler 方程 $\begin{cases} \lambda(n+1) = \lambda(n) = c \\ u(n) = -\lambda(n+1) = -c \end{cases} \Rightarrow x(n) = x(0) - nc \Rightarrow c = \frac{1}{5}$

$$u(n) = -0.2$$

离散系统极小值原理

$$\begin{cases} x(n+1) = f(x(n), u(n), n) \\ x(0) = x_0 \\ J = \varphi(x(N), N) + \sum_{n=0}^{N-1} L(x(n), u(n), n) \end{cases}$$

末端约束 $\varphi(x(N), N) = 0$; $u(n) \in \Omega$ 为容许控制域.

• 极小值原理: 最优控制满足:

(i) 哈密顿函数 $H(n) = L(x(n), u(n), n) + \lambda(n+1)f(x(n), u(n), n)$

满足正则方程
$$\begin{cases} x(n+1) = \frac{\partial H(n)}{\partial \lambda(n+1)} \\ \lambda(n) = \frac{\partial H(n)}{\partial x(n)} \end{cases}$$

(ii) 边界条件
$$\begin{cases} x(0) = x_0 \\ \varphi(x(N), N) = 0 \\ \lambda(N) = \frac{\partial \varphi(x(N), N)}{\partial x(N)} + \gamma \frac{\partial \varphi(x(N), N)}{\partial x(N)} \quad (\text{末端自由时无第二项}) \end{cases}$$

(iii) H 对 u 取极小值. (无约束时 $\frac{\partial H(n)}{\partial u(n)} = 0$)

例5.
$$\begin{cases} x(n+1) = x(n) + u(n) \\ x(0) = x_0 \\ J = \sum_{n=0}^2 (x^2(n) + u^2(n)) \end{cases}$$

求 u .

解 $H(n) = x^2(n) + u^2(n) + \lambda(n+1)(x(n) + u(n))$

$$\begin{cases} \lambda(n) = \frac{\partial H(n)}{\partial x(n)} = 2x(n) + \lambda(n+1) \\ \frac{\partial H(n)}{\partial u(n)} = 2u(n) + \lambda(n+1) = 0 \\ \lambda(3) = 0 \end{cases} \Rightarrow \begin{cases} u(n) = -\frac{1}{2}\lambda(n+1) \\ \lambda(n+1) = \lambda(n) - 2x(n) \end{cases}$$

则 $u = \{-\frac{3}{5}x_0, -\frac{1}{5}x_0, 0\}$.

• 连续系统离散化后求得的最优控制并不严格最优.

3. 特殊最优控制问题

时间最优控制

$$\begin{cases} \dot{x}(t) = f(x(t), t) + B(x(t), t)u(t) \\ \psi(x(t_f), t_f) = 0 \\ \min_{|u(t)| \leq 1} J = \int_{t_0}^{t_f} dt \end{cases}$$

取哈密顿函数 $H = 1 + \lambda(t)[f(x(t), t) + B(x(t), t)u(t)]$

$$\begin{cases} \dot{x} = f(x(t), t) + B(x(t), t)u(t) \\ \dot{\lambda} = -\frac{\partial f}{\partial x} \lambda - \frac{\partial (Bu)}{\partial x} \lambda \end{cases}$$

取 u 使 H 最小, 令 $\lambda B(x(t), t)u(t)$ 最小即可.

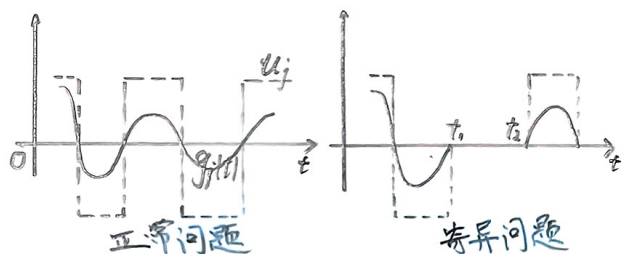
$$B(x, t) = (b_1(x, t), b_2(x, t), \dots, b_m(x, t))$$

$$g_i(x, t) = b_j(x, t) \lambda(t)$$

则 $u_j(t) = -\text{sgn}(g_j(t))$

① 奇异问题和正常问题

$$\begin{cases} \text{奇异问题: 在 } [t_1, t_2] \text{ 内存在 } g_j(t) = b_j^T(x, t) \lambda(t) = 0 \\ \text{正常问题: 在 } [t_0, t_f] \text{ 内有限时间点使 } g_j(t) = 0 \end{cases}$$



② Bang-Bang 控制: 对正常问题: $u(t) = -\text{sgn}[B(x, t)\lambda(t)]$

对奇异问题应另外寻找控制律.

线性定常系统的时间最优控制

连续系统:

$$\begin{cases} \dot{x} = Ax + Bu \\ x(0) = x_0, x(t_f) = 0 \\ \min_{|u(t)| \leq 1} J = \int_{t_0}^{t_f} dt \end{cases}$$

① 奇异问题和正常问题

$$\begin{cases} \text{奇异问题: } G_j = [b_j, Ab_j, \dots, A^{n-1}b_j] \text{ 奇异.} \\ \text{正常问题: } G_j \text{ 满秩.} \end{cases}$$

正常 $\longleftrightarrow$ 可控

例1. $\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u(t)$

$x_1(0) = x_2(0) = 1, x_1(t_f) = x_2(t_f) = 0$

$|u(t)| \leq 1$, 求时间最优控制.

解 系统能控且正常, Bang-Bang 控制时间最优.

$H = \lambda_1 x_1 + \lambda_2 u + 1$

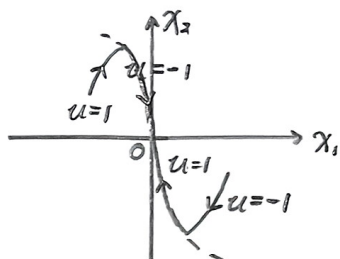
$u = \begin{cases} 1 & \lambda_2 < 0 \\ -1 & \lambda_2 > 0 \end{cases}$

由协态方程 $\begin{cases} \dot{\lambda}_1 = 0 \\ \dot{\lambda}_2 = -\lambda_1 \end{cases} \Rightarrow$ 候选控制 $\{1, -1, \{1, -1\}, \{-1, 1\}\}$

若 $u = 1$, 则 $\begin{cases} x_2(t) = x_2(0) + t \\ x_1(t) = \frac{1}{2}t^2 + x_2(0)t + x_1(0) \end{cases} \Rightarrow x_1 = \frac{1}{2}x_2^2(x_1(0) - \frac{1}{2}x_2^2(0))$

满足末态要求的相轨迹为 $\gamma_+ = \{x_1 = -\frac{1}{2}x_2^2, x_2 \leq 0\}$

若 $u = -1$, 则有 $\gamma_- = \{x_1 = -\frac{1}{2}x_2^2, x_2 \geq 0\}$



当 $x_1(0) = x_2(0) = 1$ 时, 控制序列 $\{-1, 1\}$.

$u = \begin{cases} -1 & 0 \leq t \leq 2.22 \\ 1 & 2.22 \leq t \leq 3.44 \end{cases}$

注: i) 由协态方程得到控制序列候选;

ii) 由相轨迹分析得到控制器

• 离散系统: 对 n 维离散系统 $\begin{cases} \text{若 } u \text{ 无约束, 则任意 } x(0) \text{ 在 } nT_s \text{ 内可到达原点;} \\ \text{若 } u \text{ 有约束, 则任意 } x(0) \text{ 到达原点时间大于 } nT_s. \end{cases}$

• 对于 $x(nT_s) = Ax(n) + Bu(n), x(0) = x_0, A$ 可逆且系统能控, 则由

$x(0) = GU \quad \begin{cases} G = [-A^0 & -A^{-1} & \dots & -A^{-(N-1)}]B \\ U = \begin{pmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{pmatrix} \end{cases}$

可以得到 U 使 $x(N) = 0$.

[u 无约束]

例2. $\begin{pmatrix} x_1(n+1) \\ x_2(n+1) \end{pmatrix} = \begin{pmatrix} 1 & 1-e^{-Ts} \\ e^{-Ts} & 1 \end{pmatrix} \begin{pmatrix} x_1(n) \\ x_2(n) \end{pmatrix} + \begin{pmatrix} e^{-Ts} + Ts - 1 \\ 1 - e^{-Ts} \end{pmatrix} u(n)$

$\begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} x_{10} \\ x_{20} \end{pmatrix}$, $u(n)$ 不受控制, 求 $u(n)$ 使 $x_2(2) = 0$.

解 $G = (I - A^{-1}b \quad -A^{-1}b)$

$$= - \begin{pmatrix} Ts - e^{-Ts} + 1 & Ts + e^{-Ts} - e^{2Ts} \\ e^{-Ts} - 1 & e^{2Ts} - e^{-Ts} \end{pmatrix}$$

$$= (g_0 \quad g_1)$$

由于 $\begin{cases} x_1(0) = g_0 u(0) + g_1 u(1) \\ x_1(1) = g_0 u(1) \end{cases}$, 由此有最优控制

$$u = \frac{e^{-Ts} - e^{2Ts}}{Ts(1 + e^{2Ts} - 2e^{-Ts})} x_1(n) + \frac{Ts + e^{-Ts} - e^{2Ts}}{Ts(1 + e^{2Ts} - 2e^{-Ts})} x_2(n)$$

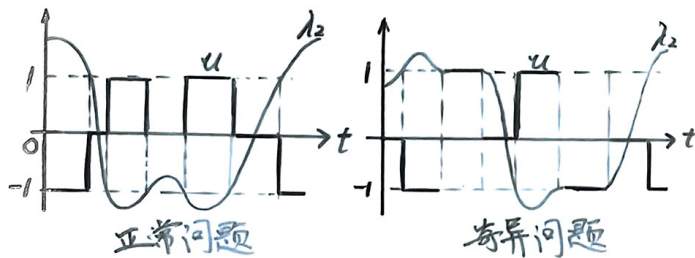
燃料最优控制 $\min J = \int_{t_0}^{t_f} \sum c_j |u_j(t)| dt$

二次积分模型的燃料最优控制 $\begin{cases} \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u \\ |u(t)| \leq 1 \\ J = \int_{t_0}^{t_f} |u(t)| dt \end{cases}$

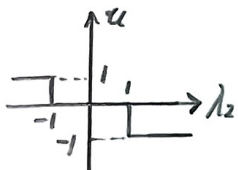
$H = |u(t)| + \lambda_1 x_2 + \lambda_2 u$

$u(t) \geq 0$ 时: $\begin{cases} u = 0 & \lambda_2 > -1 \\ u \in [0, 1] & \lambda_2 = -1 \\ u = 1 & \lambda_2 < -1 \end{cases}$ $u(t) \leq 0$ 时: $\begin{cases} u = 0 & \lambda_2 < 1 \\ u \in [-1, 0] & \lambda_2 = 1 \\ u = -1 & \lambda_2 > 1 \end{cases}$

正常问题和奇异问题: $|\lambda_2(t)| = 1$ 是否为有限点.



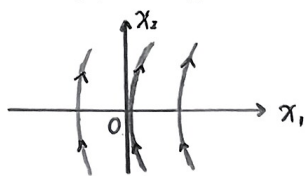
o Bang-Zero-Bang控制: 对正常问题, u 有死区特性.



o 对于二次积分模型 $\begin{cases} \dot{\lambda}_1 = c_1 \\ \dot{\lambda}_2 = -c_1 t + c_2 \end{cases} \Rightarrow \begin{cases} c_1 = 0 \text{ 时奇异} \\ c_1 \neq 0 \text{ 时不奇异 (有最多两次的控制切换)} \end{cases}$

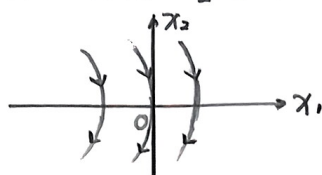
正常情况的最优控制

$$u = 1, x_1 = \frac{1}{2}x_2^2 + C$$

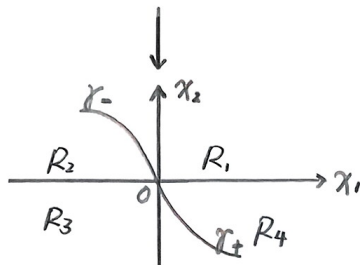


$$\gamma_+ = \{x_1 = \frac{1}{2}x_2^2, x_2 \leq 0\}$$

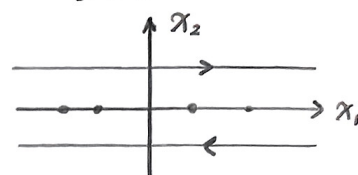
$$u = -1, x_1 = -\frac{1}{2}x_2^2 + C$$



$$\gamma_- = \{x_1 = -\frac{1}{2}x_2^2, x_2 \geq 0\}$$



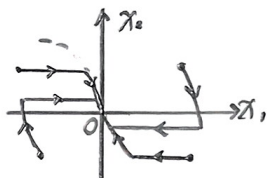
$$u = 0$$



必不最优, 若末态 $u=0$ 则无解

- (i) $(x_{10}, x_{20}) \in \gamma_+, u = 1$; (ii) $(x_{10}, x_{20}) \in R_4, u = \{0, 1\}$;
 $(x_{10}, x_{20}) \in \gamma_-, u = -1$; $(x_{10}, x_{20}) \in R_2, u = \{0, -1\}$;

(iii) $(x_{10}, x_{20}) \in R_1$ 或 R_3, u 无解, 仅有近似最优:



时间-燃料最优控制 $\min J = \rho(t_f - t_0) + \int_{t_0}^{t_f} |u(t)| dt$

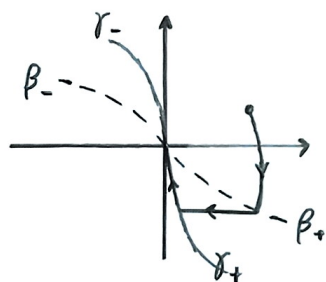
$\rho = 0 \rightarrow$ 燃料最优; $\rho \rightarrow \infty \rightarrow$ 时间最优.

二次积分模型的时间-燃料最优控制.

• 无奇异问题.

• 有两条切旋线: $\gamma = \{x_1 = -\frac{1}{2}|x_2|x_2\}$

$$\beta = \{x_1 = -\frac{\rho+4}{2\rho}|x_2|x_2\}$$



$\begin{cases} \rho \rightarrow \infty \text{ 时, } \beta_+ \text{ 与 } \gamma_- \text{ 重合, } \beta_- \text{ 与 } \gamma_+ \text{ 重合.} \\ \rho = 0 \text{ 时, } \beta_-, \beta_+ \text{ 与 } x_1 \text{ 重合.} \end{cases}$

第三章 线性二次型问题

1. 线性二次型问题

- 线性二次型问题：系统为线性系统，性能指标为 u, x 的二次型函数。
 - 线性最优控制结果可用于小信号非线性控制。
 - 线性二次型最优控制有良好的频率响应。

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$u(t)$ 无约束, y_r 为期望输出, 误差 $e(t) = y_r(t) - y(t)$

$$J = \frac{1}{2} e^T(t_f) F e(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (e^T(t) Q e(t) + u^T(t) R u(t)) dt$$

↓	↓	↓
末态误差	动态跟踪误差	控制能量消耗
F : 半正定常阵	Q : 对称时变阵	R : 正定对称时变阵

状态调节器: $J = \frac{1}{2} x^T(t_f) F x(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (x^T Q x + u^T R u) dt$
($C(t) = E, y_r(t) = 0$) 产生 u 使 x 在平衡状态附近。

输出调节器: $J = \frac{1}{2} y^T(t_f) F y(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (y^T Q y + u^T R u) dt$
($y_r(t) = 0$) 产生 u 使 y 在平衡状态附近。

跟踪系统
($y_r(t) \neq 0$)

2. 状态调节器 (LQR Controller)

连续系统状态调节器

有限时间调节器

$$\begin{cases} \dot{x} = Ax + Bu \\ x(t_0) = x_0 \\ J = \frac{1}{2} x^T(t_f) F x(t_f) + \frac{1}{2} \int_{t_0}^{t_f} [x^T(t) Q x(t) + u^T(t) R u(t)] dt \\ F = F^T \geq 0; Q = Q^T \geq 0; R = R^T > 0. \end{cases}$$

最优控制

$$\begin{cases} \text{控制律 } u(t) = -R^{-1}(t)B^T(t)P(t)x(t) \\ \text{Riccati 方程 } \begin{cases} -\dot{P}(t) = P(t)A(t) + A^T(t)P(t) - P(t)B(t)R^{-1}(t)B^T(t)P(t) + Q(t) \\ P(t_f) = F \end{cases} \end{cases}$$

$$\begin{cases} \text{最优性能指标: } J_{\min} = \frac{1}{2} x^T(t_0) P(t_0) x(t_0) \\ \dot{x} = (A - BR^{-1}B^T P)x \end{cases}$$

$$H = \frac{1}{2} x^T Q x + \frac{1}{2} u^T R u + \lambda^T A x + \lambda^T B u$$

由于 u 无约束, 则由极小值原理, $\frac{\partial H}{\partial u} = Ru + B^T \lambda = 0 \Rightarrow u = -R^{-1}B^T \lambda$

由于 $\frac{\partial^2 H}{\partial u^2} = R > 0$, 为最优控制

$$\text{正则方程 } \begin{cases} \dot{x} = Ax - BR^{-1}B^T \lambda, \text{ 若令 } \lambda = Px, \dot{\lambda} = \dot{P}x + P\dot{x} \\ \dot{\lambda} = -Qx - A^T \lambda \end{cases}$$

$$\begin{aligned} \text{由 } \dot{x} = Ax - BR^{-1}B^T \lambda &\Rightarrow \dot{x} = (A - BR^{-1}B^T P)x \\ &\Rightarrow \begin{cases} \dot{\lambda} = -(Q + A^T P)x \\ \dot{\lambda} = (\dot{P} + PA - PBR^{-1}B^T P)x \end{cases} \Rightarrow \text{Riccati 方程} \end{aligned}$$

① u 为线性状态反馈: $K = R^{-1}B^T P$

ii) P 是唯一对称非负矩阵;

iii) P 是时变阵, 实现困难.

$$\text{例1. } \begin{cases} \dot{x} = \frac{1}{2}x + u \\ x(t_0) = x_0 \\ J = \int_{t_0}^{t_f} (\frac{1}{2}e^{-t}x^2 + 2e^{-t}u^2) dt \end{cases}$$

$$\text{解 } A = \frac{1}{2}, B = 1, F = 0, Q = \frac{1}{2}e^{-t}, R = 2e^{-t}.$$

$$\text{由 Riccati 方程求 } P(t): \begin{cases} -\dot{P} = P - \frac{1}{2}e^t P^2 + \frac{1}{2}e^{-t} \\ P(t_f) = 0 \end{cases}$$

为使方程线性化, 令 $\hat{x} = e^{-\frac{1}{2}t}x, \hat{u} = e^{-\frac{1}{2}t}u$

$$\begin{cases} \dot{\hat{x}} = \hat{u} \\ -\dot{\hat{p}}(t) = -\frac{1}{2}\hat{p}^2(t) + \frac{1}{2}, \quad \hat{p}(t_f) = 0 \end{cases} \Rightarrow \begin{cases} \hat{p} = \frac{1 - e^{t-t_f}}{1 + e^{t-t_f}} \\ \hat{u} = -\frac{1}{2}e^{-\frac{1}{2}t}\hat{p}x \end{cases}$$

$$\Rightarrow u = -\frac{1}{2} \frac{1 - e^{t-t_f}}{1 + e^{t-t_f}} x$$

无限时间调节器 有限时间调节器只考虑恢复到平衡态;
无限时间调节器还考虑保持平衡态.

$$\begin{cases} \dot{x} = Ax + Bu \\ x(t_0) = x_0 \\ J = \frac{1}{2} \int_{t_0}^{\infty} (x^T Q x + u^T R u) dt \end{cases}$$

四 时变调节器

· 最优控制: 若 $\{A, B\}$ 完全能控, 最优控制存在:

$$u = -R^{-1}B^T \bar{P} x$$

$$\text{Riccati 方程} \begin{cases} -\dot{\bar{P}} = \bar{P}A + A^T \bar{P} - \bar{P}B R^{-1} B^T \bar{P} + Q, \quad \bar{P}(t_f) = 0 \\ \bar{P} = \lim_{t_f \rightarrow \infty} \bar{P} \end{cases}$$

ii) $t_f \rightarrow \infty$ 时, 要求系统能控.

$F=0$, 因为 $t_f \rightarrow \infty$ 时末态指标无意义.

四 定常调节器

· 最优控制: 若 $\{A, B\}$ 完全能控, $\{A, D\}$ 完全能观, 最优控制存在:

$$\downarrow$$

$$DD^T = Q, D \text{ 任意.}$$

$$u = -R^{-1}B^T \bar{P} x$$

Riccati 方程: $\bar{P}A + A^T \bar{P} - \bar{P}B R^{-1} B^T \bar{P} + Q = 0$ (即 $\dot{\bar{P}} = 0$, $\bar{P}$ 为常阵).

且该系统稳定 ($\{A, D\}$ 可观得到).

① 给定稳定度 (极点小于 $-\alpha$): $J = \frac{1}{2} \int_{t_0}^{\infty} e^{2\alpha t} (x^T Q x + u^T R u) dt$

此时 Riccati 方程: $\bar{P}(A + \alpha E) + (A + \alpha E)^T \bar{P} - \bar{P}B R^{-1} B^T \bar{P} + Q = 0$.

离散系统状态调节器

$$\begin{cases} x(n+1) = A(n)x(n) + B(n)u(n) \\ J = \frac{1}{2} x^T(N) F x(N) + \frac{1}{2} \sum_{n=0}^N (x^T Q x + u^T R u) \\ F \geq 0; Q(n) \geq 0; R(n) > 0. \end{cases}$$

· 最优控制: 和连续系统类似.

$$u(n) = -K(n)x(n)$$

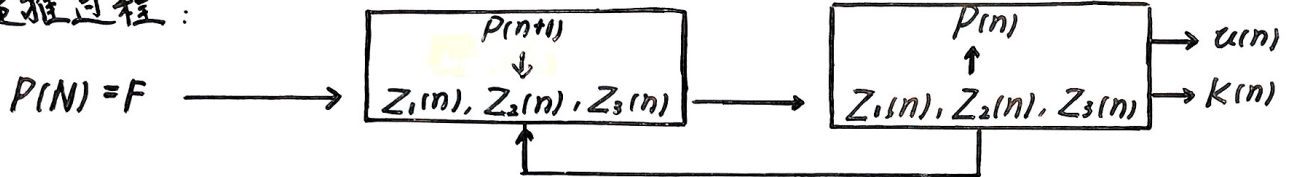
$$\text{反馈阵: } K(n) = (B^T(n)P(n+1)B(n) + R(n))^{-1}B^T(n)P(n+1)A(n)$$

$$\text{Riccati方程} \begin{cases} P(n) = [A(n) - B(n)K(n)]^T P(n+1) [A(n) - B(n)K(n)] + K^T(n)R(n)K(n) + Q(n) \\ P(N) = F \end{cases}$$

◎ Riccati 方程可离线求解：逆向递推

$$\text{令} \begin{cases} Z_1(n) = Q(n) + A^T(n)P(n+1)A(n) \\ Z_2(n) = B^T(n)P(n+1)A(n) \\ Z_3(n) = R(n) + B^T(n)P(n+1)B(n) \end{cases} \Rightarrow \begin{cases} K(n) = Z_3^{-1}(n)Z_2(n) \\ P(n) = Z_1(n) - Z_2^T(n)Z_3^{-1}(n)Z_2(n) \end{cases}$$

递推过程：



◎ $N \rightarrow \infty$ 时, 取 $P(n) = P(n+1)$, 退化为代数 Riccati 方程.

3. 输出调节器与跟踪系统

输出调节器

有限时间输出调节器

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \\ J = \frac{1}{2} y^T(t_f) F y(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (y^T Q y + u^T R u) dt \\ F \geq 0; Q \geq 0; R > 0. \end{cases}$$

• 最优控制: $\{A, C\}$ 能观时有最优控制:

$$u(t) = -R^{-1} B^T P x(t)$$

$$\text{Riccati 方程} \begin{cases} -\dot{P} = PA + A^T P - P B R^{-1} B^T P + C^T Q C \\ P(t_f) = C^T(t_f) F C(t_f) \end{cases}$$

无限时间定常输出调节器

• 最优控制: $\{A, B\}$ 能控, $\{A, C\}$ 能观, $\{A, D\}$ 能观, 有最优控制

$$PD^T = C^T Q C$$

$$u(t) = -R^{-1} B^T \bar{P} x(t)$$

$$\text{Riccati 方程: } 0 = \bar{P} A + A^T \bar{P} - \bar{P} B R^{-1} B^T \bar{P} + C^T Q C$$

输出反馈调节器 $\rightarrow$ 使用输出反馈, 次优 (无限时间)

• 次优控制: (i) 令 $\begin{cases} \bar{A} = A + B K C \\ \bar{a} = a + C^T K^T R K C \\ u = -K y \end{cases}$

(ii) K 满足 Lyapunov 方程:

$$\bar{A}^T P + P \bar{A} = -\bar{Q} \quad (\text{无法直接解})$$

(iii) 解出 $P(k)$, 令 $J(k) = \frac{1}{2} x^T(0) P x(0)$, $\frac{\partial J(k)}{\partial k} = 0$, 可得 k .

(k 与 $x(0)$ 有关)

离散输出调节器

$$\begin{cases} x(n+1) = A(n)x(n) + B(n)u(n) \\ y(n) = C(n)x(n) \\ J = \frac{1}{2} y^T(N) F y(N) + \frac{1}{2} \sum_{n=0}^N (y^T Q y + u^T R u) \end{cases}$$

• 最优控制: $\{A, C\}$ 能观时有最优控制:

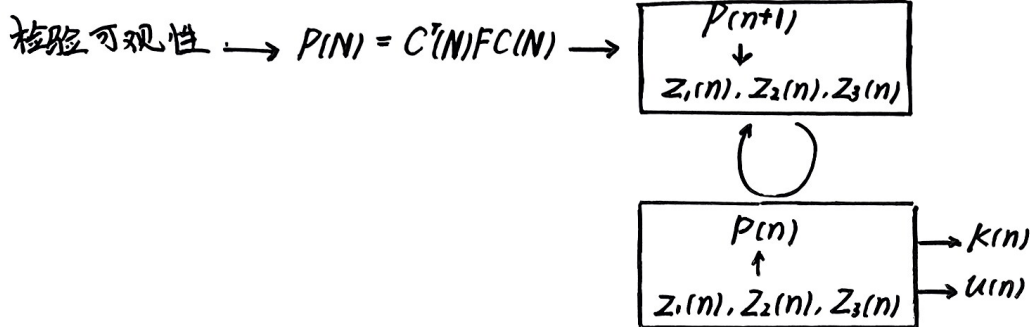
$$u(n) = -K(n)u(n)$$

$$\text{Riccati 方程} \begin{cases} P(n) = [A(n) - B(n)K(n)]^T P(n+1) [A(n) - B(n)K(n)] + K^T(n) R(n) K(n) + C^T(n) Q(n) C(n) \\ P(N) = C^T(N) F C(N) \end{cases}$$

• 离线求解器:

$$\begin{cases} Z_1(n) = C^T(n)Q(n)C(n) + A^T(n)P(n+1)A(n) \\ Z_2(n) = B^T(n)P(n+1)A(n) \\ Z_3(n) = R(n) + B^T(n)P(n+1)B(n) \end{cases}$$

$$\Rightarrow \begin{cases} K(n) = Z_2^{-1}(n) Z_1(n) \\ P(n) = Z_1(n) - Z_2^T(n) Z_3^{-1}(n) Z_2(n) \end{cases}$$



跟踪系统

有限时间跟踪系统

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \\ e(t) = y_d(t) - y(t), y_d(t) \text{ 为期望输出.} \\ J = \frac{1}{2} e^T(t_f) F e(t_f) + \frac{1}{2} \int_{t_0}^{t_f} (e^T Q e + u^T R u) dt. \end{cases}$$

• 最优控制: $[A, C]$ 可观, 最优控制存在.

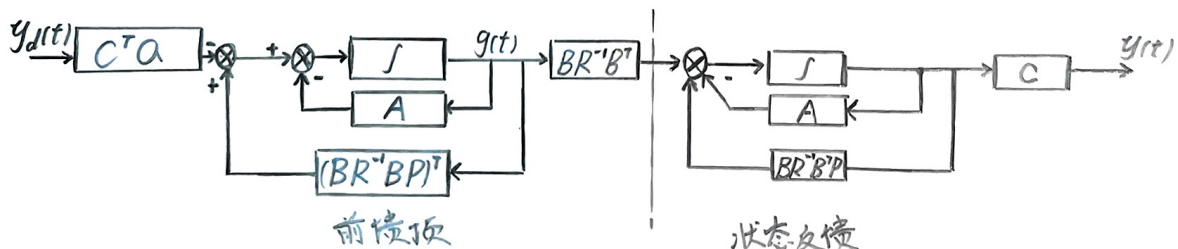
$$u = -R^{-1}B^T(Px - g)$$

Riccati 方程 $\begin{cases} -\dot{P} = PA + A^T P - PBR^{-1}B^T P + C^T Q C \\ P(t_f) = C^T(t_f) F C(t_f) \end{cases}$

伴随向量 $g \begin{cases} -\dot{g} = (A - BR^{-1}B^T P)g + C^T Q C \\ g(t_f) = C^T(t_f) F y(t_f) \end{cases}$

① 反馈结构和期望输出无关: 增加前馈 $R^{-1}B^T g$.

↓
动态性能和期望输出无关.



② 一般取 $y_d(t)$ 视为某典型变化律/随机信号

↓
性能取决于 $y(t)$ 与 $y_d(t)$ 符合程度

↓
性能平均意义最优.

无限时间定常跟踪系统 $y_d(t)$ 为期望常值.

· 近似最优控制: $\{A, B\}$ 能控, $\{A, C\}$ 能观.

$$u = -R^{-1}B^T P x + R^{-1}B^T g$$

Riccati 方程: $PA + A^T P - PBR^{-1}B^T P + C^T Q C = 0$

伴随向量: $g = (PBR^{-1}B^T - A^T)^{-1} C^T Q y_d$

2. 模型预测控制

预测方程

$$\begin{cases} x(n+1) = Ax(n) + B_u u(n) + B_d d(n) \\ y(n) = Cx(n) \end{cases}$$

改写为增量式:
$$\begin{cases} \Delta x(n+1) = A\Delta x(n) + B_u \Delta u(n) + B_d \Delta d(n) \\ y(n) = C\Delta x(n) + y(n-1) \end{cases}$$

$$\Delta: [n] - [n-1]$$

预测步长为 p , 控制步长为 m , $m \leq p$

假设: (i) 控制步长外控制量不变: $\Delta u(n+i) = 0, i = m, \dots, p-1$

(ii) 干扰在 n 时刻后不变: $\Delta d(n+i) = 0, i = 1, \dots, p-1$

$$\Delta x(n+1|n) = A\Delta x(n) + B_u \Delta u(n) + B_d \Delta d(n)$$

$$\Delta x(n+2|n) = A\Delta x(n+1|n) + B_u \Delta u(n+1) + B_d \Delta d(n+1)$$

$$= A^2 \Delta x(n) + AB_u \Delta u(n) + B_u \Delta u(n+1) + AB_d \Delta d(n)$$

$$\Delta x(n+m|n) = A^m \Delta x(n) + A^{m-1} B_u \Delta u(n) + A^{m-2} B_u \Delta u(n+1) + \dots + B_u \Delta u(n+m-1) + A^{m-1} B_d \Delta d(n)$$

⋮

$$\Delta x(n+p|n) = A^p \Delta x(n) + A^{p-1} B_u \Delta u(n) + A^{p-2} B_u \Delta u(n+1) + \dots + A^{p-m} B_u \Delta u(n+m-1) + A^{p-1} B_d \Delta d(n)$$

↓

$$y(n+1|n) = CA\Delta x(n+1|n) + y(n)$$

$$= CA\Delta x(n) + CB_u \Delta u(n) + CB_d \Delta d(n) + y(n)$$

⋮

$$y(n+m|n) = \sum_{i=0}^{m-1} CA^i \Delta x(n) + \sum_{i=0}^{m-1} CA^{i-1} B_u \Delta u(n) + \sum_{i=0}^{m-1} CA^{i-1} B_u \Delta u(n+1) + \dots + CB_u \Delta u(n+m-1) + \sum_{i=0}^{m-1} CA^{i-1} B_d \Delta d(n) + y(n)$$

• 预测方程: 定义

$$Y_p(n+1|n) = \begin{pmatrix} y(n+1|n) \\ y(n+2|n) \\ \vdots \\ y(n+p|n) \end{pmatrix}$$

↓
预测输出向量

$$\Delta U(n) = \begin{pmatrix} \Delta u(n) \\ \Delta u(n+1) \\ \vdots \\ \Delta u(n+m-1) \end{pmatrix}$$

↓
输入向量

$$S_x = \begin{pmatrix} CA \\ \sum_{i=1}^p CA^i \\ \vdots \\ \sum_{i=1}^p CA^i \end{pmatrix}, S_d = \begin{pmatrix} CB_d \\ \sum_{i=1}^p CA^{i-1} B_d \\ \vdots \\ \sum_{i=1}^p CA^{i-1} B_d \end{pmatrix}, S_u = \begin{pmatrix} CB_u & 0 & \dots & 0 \\ \sum_{i=1}^m CA^{i-1} B_u & CB_u & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^m CA^{i-1} B_u & \sum_{i=1}^{m-1} CA^{i-1} B_u & \dots & CB_u \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^p CA^{i-1} B_u & \sum_{i=1}^{p-1} CA^{i-1} B_u & \dots & \sum_{i=1}^{p-m} CA^{i-1} B_u \end{pmatrix}$$

$$Y_p(n+1|n) = S_x \Delta x(n) + E y_c(n) + S_d \Delta d(n) + S_u \Delta u(n)$$

- 假设：
 (i) 控制步长之外的控制量不变；
 (ii) 干扰在预测步长内不变。

无约束预测控制

优化问题数学描述 采用二次型代价：

$$J = \sum_{i=1}^p \|Q[y(n+i|n) - y_d(n+i)]\|^2 + \sum_{i=1}^m \|R[\Delta u(n+i-1)]\|^2$$

Q, R 为对角矩阵，表征对输出误差和控制输出的权重。

越大误差越小 越大控制动作越小。

↓

$$J = \|Q[Y_p(n+1|n) - Y_d(n+1)]\|^2 + \|R\Delta u(n)\|^2$$

优化问题的解

$$\rho \stackrel{\text{def}}{=} \begin{pmatrix} Q[Y_p(n+1|n) - Y_d(n+1)] \\ R[\Delta u(n)] \end{pmatrix} \Rightarrow J = \rho^T \rho$$

代入预测方程：

$$\rho = \underbrace{\begin{pmatrix} Q S_u \\ R \end{pmatrix} \Delta u(n)}_{Az} - \underbrace{\begin{pmatrix} Q [Y_d(n+1) - S_x \Delta x(n) - E y(n) - S_d \Delta d(n)] \\ 0 \end{pmatrix}}_{b}$$

$E_p(n+1|n)$

$$J = \rho^T \rho \text{ 极值为 } z^* = (A^T A)^{-1} A^T b$$

• 最优控制序列：

$$\Delta u(n) = (S_u^T Q^T A S_u + R^T R)^{-1} S_u^T Q^T A E_p(n+1|n)$$

↓

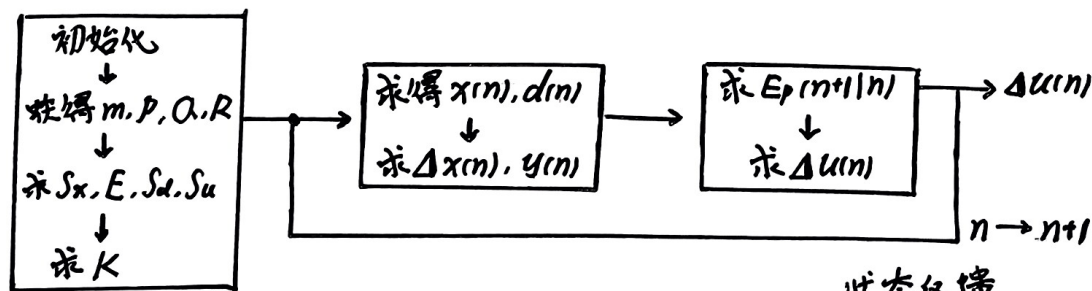
$$Y_d(n+1) - S_x \Delta x(n) - E y(n) - S_d \Delta d(n)$$

↓

MPC 控制器给出的控制： $\Delta u(n) = K E_p(n+1|n)$

K 为控制增益，可离线计算。

• MPC 算法。



• 前馈-反馈特点：
$$\Delta u(n) = \underbrace{K Y_d(n+1)}_{\text{前馈}} - \underbrace{K(S_x + EC)x(n)}_{\text{反馈}} - \underbrace{K S_d \Delta d(n)}_{\text{前馈}} + \underbrace{K S_x x(n-1)}_{\text{反馈}}$$

• MPC 控制的性能

① 稳定性: $A - B_u K(S_x + EC)$ 的特征值应均在单位圆内.

② 抗干扰: 当外部干扰趋于不变时, 状态和输出有界.

③ MPC 无静差.

基于估计的控制 系统状态不全可测或有测量噪声时, 将使用状态观测器

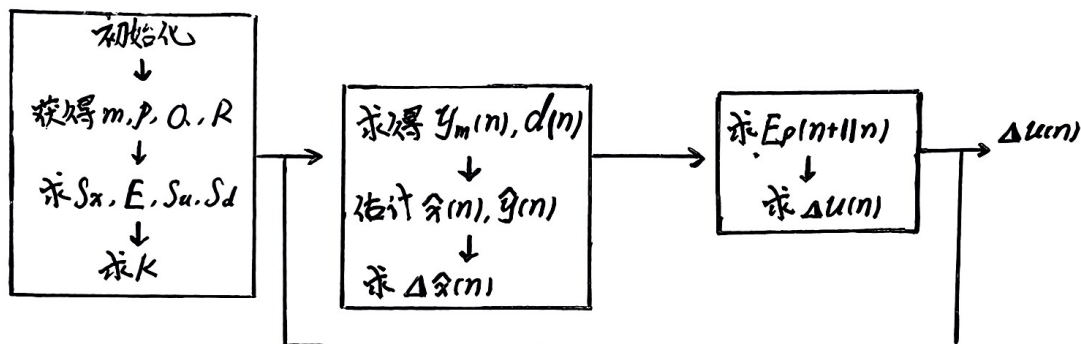
$$\begin{cases} \hat{x}(n+1) = A\hat{x}(n) + B_u u(n) + B_d d(n) + \frac{L[y_m(n) - C_m \hat{x}(n)]}{\text{观测器的反馈}} \\ \hat{y}(n) = C\hat{x}(n) \end{cases}$$

C_m 可测量 (不可测的状态稳定) 时, 可用观测器设计 L 使 $A - LC$ 稳定.

• ④ 预测方程: $Y_p(n+1|n) = S_x \Delta \hat{x}(n) + E \hat{y}(n) + S_d \Delta d(n) + S_u \Delta u(n)$

⑤ 控制律

$$\begin{cases} E_p(n+1|n) = Y_d(n+1) - S_x \Delta \hat{x}(n) - E \hat{y}(n) - S_d \Delta d(n) \\ \Delta u(n) = K E_p(n+1|n) \end{cases}$$



有约束预测控制

$$\begin{cases} \Delta x(n+1) = A \Delta x(n) + B_u \Delta u(n) + B_d \Delta d(n) \\ y(n) = C \Delta x(n) + y(n-1) \\ y_b(n) = C_b \Delta x(n) + y_b(n-1) \longrightarrow \text{约束输出} \end{cases}$$

$$u_{\min}(n) \leq u(n) \leq u_{\max}(n); \Delta u_{\min}(n) \leq \Delta u(n) \leq \Delta u_{\max}(n); y_{\min}(n) \leq y_b(n) \leq y_{\max}(n)$$

• 预测方程: $Y_p(n+1|n) = S_x \Delta x(n) + E y(n) + S_d \Delta d(n) + S_u \Delta u(n)$

约束优化求解

④ 目标函数转换

$$\begin{aligned} J &= \| \alpha [S_u \Delta u(n) - E_p(n+1|n)] \|^2 + \| R \Delta u(n) \|^2 \\ &= \Delta u(n)^T S_u^T \alpha^T \alpha S_u \Delta u(n) + \Delta u(n)^T R^T R \Delta u(n) \\ &\quad - 2 E_p(n+1|n) \alpha^T \alpha S_u \Delta u(n) + E_p(n+1|n) \alpha^T \alpha E_p(n+1|n) \\ &\Downarrow \\ \tilde{J} &= \Delta u(n)^T H \Delta u(n) + G^T \Delta u(n) \end{aligned}$$

③ 控制约束转换

$$\begin{pmatrix} -T \\ T \end{pmatrix} \Delta U(n) \geq \begin{pmatrix} -\Delta u_{\max}(n) \\ \vdots \\ -\Delta u_{\max}(n+m-1) \\ \Delta u_{\min}(n) \\ \vdots \\ \Delta u_{\min}(n+m-1) \end{pmatrix}$$

$$T = \begin{pmatrix} E & & \\ & E & \\ & & \ddots \\ & & & E \end{pmatrix}$$

$$\begin{pmatrix} -L \\ L \end{pmatrix} \Delta U(n) \geq \begin{pmatrix} u(n-1) - u_{\max}(n) \\ \vdots \\ u(n-1) - u_{\max}(n+m-1) \\ u_{\min}(n) - u(n-1) \\ \vdots \\ u_{\min}(n+m-1) - u(n-1) \end{pmatrix}$$

$$L = \begin{pmatrix} E & & & \\ E & E & & \\ \vdots & \vdots & \ddots & \\ E & E & \dots & E \end{pmatrix}$$

④ 输出约束转换

$$\text{约束输出: } Y_{pb}(n+1|n) = S_{xb} \Delta x(n) + E_b y_b(n) + S_{db} \Delta d(n) + S_{db} \Delta d(n)$$

↓

$$\begin{pmatrix} -S_{ub} \\ S_{ub} \end{pmatrix} \Delta U(n) \geq \begin{pmatrix} S_{xb} \Delta x(n) + E_b y_b(n) + S_{db} \Delta d(n) - Y_{\max}(n+1) \\ -S_{xb} \Delta x(n) - E_b y_b(n) - S_{db} \Delta d(n) + Y_{\min}(n+1) \end{pmatrix}$$

$$\text{ap问题} \begin{cases} \min J = \Delta U^T(n) H \Delta U(n) - G \Delta U(n) \\ C_u \Delta U(n) \geq b. \end{cases}$$